Inventor Mobility, Knowledge Diffusion, and Growth

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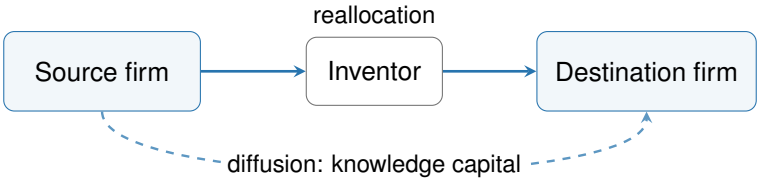
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The views expressed here are those of the authors and do not necessarily reflect the views of the Bank of Japan.

How does inventor mobility affect growth and welfare?

- "Mobility of personnel among firms provides a way of spreading information" (Arrow, 1962)



- Non-compete clauses affect inventor mobility, as in Silicon Valley against Route 128 (Saxenian, 1994; Gilson, 1999)
- German inventors' annual job-to-job rate fell from 12.5% in 2001 to 4.8% in 2014

\+ Inventors move to firms that value them more (reallocation)

\+ The destination firm absorbs the source firm's knowledge (diffusion)

– Search is costly, and search and innovation externalities distort mobility

Motivating Facts and the Model

- German employer–employee matched data linked to patent data
- Patent-listed inventors often move to lower-mean-wage establishments
- Inventors move down more often than other workers yet receive larger wage gains
- A firm-wage job ladder, in which moves go to higher-wage employers, generates neither fact
- An endogenous growth model with on-the-job search, in which a move transfers knowledge
- That knowledge makes moves to less productive firms profitable

Main Results

- A non-compete ban raises inventor mobility and the annual growth rate
- The 2001–2014 mobility decline lowers the annual growth rate by 0.25 pp
- Knowledge diffusion, not reallocation, drives most of both growth effects
- A non-compete ban lowers consumption-equivalent (CE) welfare by 0.41%

Literature

- Knowledge diffusion & growth
  - Kortum (1997); Lucas and Moll (2014); Perla and Tonetti (2014); Akcigit et al. (2018); Buera and Oberfield (2020); Benhabib et al. (2021); Prato (2025); Crews (2023); Babalievsky (2023)
  - Job-to-job transition $\rightarrow$ knowledge diffusion
- Inventor mobility & knowledge diffusion
  - Almeida and Kogut (1999), Song et al. (2003), Rosenkopf and Almeida (2003), Hoisl (2007), Breschi and Lissoni (2009), Singh and Agrawal (2011), Kaiser et al. (2015), Rahko (2017), and Braunerhjelm et al. (2020)
  - Quantifying growth effects and policy counterfactuals
- Non-compete clauses
  - Marx et al. (2009); Balasubramanian et al. (2022); Shi (2023); Jeffers (2024); Baslandze and Vardishvili (2026); Gottfries and Jarosch (2026)
  - Moves between incumbent firms that transfer knowledge

Motivating Facts

German Inventor and Worker Data

- **INV-BIO.** 152,350 inventors named on EPO patent applications 1999–2011, with their social-security employment and wage histories 1980–2014
- **SIAB.** A 2% random sample of the same records (1,940,962 individuals), without patents
- The two data sets cannot be linked

Fact 1-1: Many inventor moves go to lower-wage establishments

Inventor moves by establishment mean-wage bin (row %)

	Destination rank bin									
Origin	1	2	3	4	5	6	7	8	9	10
1	18	16	13	12	10	9	7	6	5	4
2	11	16	14	12	10	9	8	8	7	6
3	7	13	14	12	10	9	9	9	8	8
4	6	10	11	12	11	11	11	10	10	8
5	5	8	9	11	11	11	10	12	11	12
6	4	7	9	9	10	12	13	12	12	11
7	4	6	7	8	9	11	13	13	14	15
8	3	5	6	7	10	11	14	13	14	17
9	2	5	5	6	9	9	14	14	16	21
10	2	3	4	4	6	7	10	12	19	32

- Ranked by mean wage, 36% of inventor moves go to a lower bin

INV-BIO, 1980–2014, inventor job-to-job moves. Establishments ranked within year by previous-year mean wage, in ten bins formed by cumulative inventor counts.

Moves with wage gains

Fact 1-2: Many inventor moves go to less productive establishments

Inventor moves by establishment citation-productivity bin (row %)

	Destination rank bin									
Origin	1	2	3	4	5	6	7	8	9	10
1	12	12	11	10	8	9	9	9	10	10
2	12	12	12	9	9	9	9	9	10	10
3	10	12	12	11	9	10	9	9	9	9
4	11	10	11	11	11	10	11	9	9	8
5	9	9	10	9	10	11	15	10	9	10
6	8	8	10	7	12	9	9	12	11	13
7	6	8	6	14	12	9	12	11	11	12
8	5	6	7	7	8	12	12	14	14	14
9	4	5	10	6	12	8	9	16	13	17
10	5	7	6	6	6	10	10	11	15	25

- Ranked by citation productivity, 37% of inventor moves go to a lower bin

INV-BIO, 1980–2014, inventor job-to-job moves. Establishments ranked within year by the three-year backward average of citation productivity per inventor (positive values only), in ten bins formed by cumulative inventor counts.

Fact 2: Inventor-proxy movers move down more often yet earn more

	Move to lower-mean-wage establishment	Own-wage growth
Inventor proxy $I_{i,t-1}$	0.0080^{***}	0.0581^{***}
	(0.0021)	(0.0033)

SIAB job-to-job movers, $N = 651{,}554$. Standard errors two-way clustered by origin and destination establishment.

- Inventor proxies are workers with a university-entrance qualification in the five occupation groups most common among inventors
- Controls: age, age squared, gender, log wage; origin, destination, and year FE
- A firm-wage job ladder, in which moves go to higher-wage employers, generates neither fact

Model

Heterogeneous Firms with Knowledge Capital and Inventors

- Continuum of firms with
  1. continuum mass of knowledge capital (or product lines) k
  2. continuum mass of inventors n
- Fixed total mass of inventors N
- Instantaneous payoff

$$\pi k - c\left(\frac{\hat{v}}{k}\right)k$$

  - π: profit per product line
  - $\hat{v}$: vacancy to search inventors
  - $c(\cdot)$: increasing and convex

$dk =$ Internal R&D $+$ Knowledge Diffusion

1. Internal R&D ($-$ Destruction)

$$dk|_{\text{R\&D}} = \left(\underbrace{\mu(n/k)\,k}_{\text{Internal R\&D}} - \underbrace{\delta k}_{\text{Destruction}} \right) dt + \underbrace{\sigma k\, dW}_{\text{Shocks}}$$

   - $\mu(\cdot)$: non-negative, increasing, and concave
   - δ: determined in GE so that the total mass of product lines stays at one
   - Reminiscent of Klette and Kortum (2004)
   - $\underbrace{\mu(n/k)\,k/n}_{\text{R\&D outcome per inventor}}$ is increasing in $k/n \to k/n$: inventor productivity

2. Knowledge diffusion through inventor moves
   - When inventors move from a (k', n')-firm to a (k, n)-firm, the hiring firm gains $\alpha(k'/n')$ units of knowledge capital per inventor: $dk/dn|_{(k',n')\to(k,n)} = \alpha(k'/n')$
   - The source firm keeps its knowledge capital

Matching Technology

- Random search
- Only on-the-job search
- Labor market tightness $\theta = (\text{Total mass of vacancies } V) / (\text{Total mass of inventors } N)$
- A vacancy meets an inventor at rate $q(\theta)$
- An inventor meets a vacancy at rate $\theta q(\theta)$

Bargaining and Non-Compete Enforcement

Assumption.

1. In a meeting, the match is formed if the match surplus is positive
2. Poaching firm receives share $(1-\beta)$ of the match surplus, and the poached firm and its inventor receive share β
3. Vacancy posting decision maximizes the joint value of firm and incumbent inventors

- Without an enforceable non-compete clause, Bertrand competition gives $\beta = 0$ (Postel-Vinay and Robin, 2002)
- A non-compete clause secures $\beta > 0$ because the poaching firm must buy out the clause (Shi, 2023)
- Assumptions follow Lentz and Mortensen (2022); Bilal et al. (2022) microfound assumption 3
- Decisions maximize the joint value of the firm and its inventors, so the firm's state is (k, n), not the distribution of wage contracts

HJB Equation

$\Omega(k,n,t)$: joint value of n inventors and firm with knowledge capital k

$$r(t)\Omega(k,n,t) - \frac{\partial}{\partial t}\Omega(k,n,t) =$$

$$\max_{\hat{v}\geq 0} \underbrace{\pi k - c\left(\frac{\hat{v}}{k}\right)k}_{\text{Profit}-\text{Search cost}} + \underbrace{\{\mu(n/k) - \delta\}k\Omega_k(k,n,t) + \frac{\sigma^2}{2}k^2\Omega_{kk}(k,n,t)}_{\text{Internal R\&D}-\text{Destruction}}$$

$$+ \underbrace{(1-\beta)q(\theta)\hat{v}\int \hat{M}(k,n,k',n',t)dG_n(k',n',t)}_{\text{Poaching}} + \underbrace{\beta\theta q(\theta)n\int \hat{M}(k',n',k,n,t)dG_v(k',n',t)}_{\text{Being poached}}$$

$$\underbrace{\hat{M}(k,n,k',n',t)}_{\text{Match surplus}} = \Big[\underbrace{\Omega_n(k,n,t)}_{\substack{\text{destination}\\ \text{inventor value}}} + \underbrace{\alpha(k'/n')\Omega_k(k,n,t)}_{\text{value of the knowledge brought}} - \underbrace{\Omega_n(k',n',t)}_{\substack{\text{source}\\ \text{inventor value}}}\Big]^+$$

- $G_n(k',n',t)$: inventor-weighted firm distribution
- $G_v(k',n',t)$: vacancy-weighted firm distribution

A continuum of product lines makes $x = n/k$ a sufficient state variable

- $x \equiv n/k$ is inventors per product line; $x^{-1} = k/n$ is inventor productivity
- Klette and Kortum (2004): with discrete product lines, the variance of firm growth falls with size, so both k and n are state variables
- Here, with a continuum of product lines and proportional shocks, $d \log k$ and $d \log n$ depend only on x, and $\Omega(k, n) = k\, S(x)$
- Firms never lose all product lines, so the model abstracts from entry, exit, and unemployment

Simplified HJB Equation

In the steady state, the joint value per product line $S(x)$ satisfies

$$\rho S(x) = \max_{v \geq 0} \underbrace{\pi - c(v)}_{\text{Profit}-\text{Search cost}} + \underbrace{\{\mu(x) - \delta\}\{S(x) - xS_x(x)\} + \frac{\sigma^2}{2}x^2 S_{xx}(x)}_{\text{Internal R\&D}-\text{Destruction}}$$

$$+ \underbrace{(1-\beta)q(\theta)v\int M(x,x')dF_x(x')}_{\text{Poaching}} + \underbrace{\beta\theta q(\theta)x\int M(x',x)dF_v(x')}_{\text{Being poached}}$$

$$\underbrace{M(x,x')}_{\text{Match surplus}} = \left[\ S_x(x) + \alpha(x')\{S(x) - xS_x(x)\} - S_x(x')\ \right]^+$$

- v: vacancy per product line; $F(x)$: distribution of product lines over x
- $F_x(x)$, $F_v(x)$: inventor-weighted and vacancy-weighted product line distributions

Knowledge makes downward moves profitable

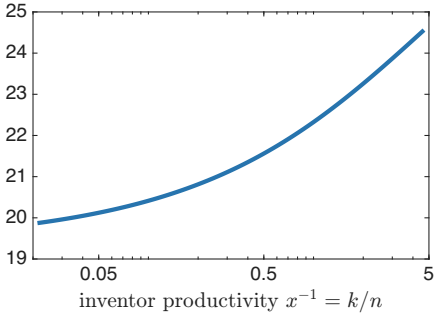
- A move from x' to x has match surplus

$$M(x,x') = \Big[\underbrace{S_x(x)}_{\substack{\text{destination}\\ \text{inventor value}}} + \underbrace{\alpha(x')\{S(x) - xS_x(x)\}}_{\text{value of the knowledge brought}} - \underbrace{S_x(x')}_{\substack{\text{source}\\ \text{inventor value}}} \Big]^+$$

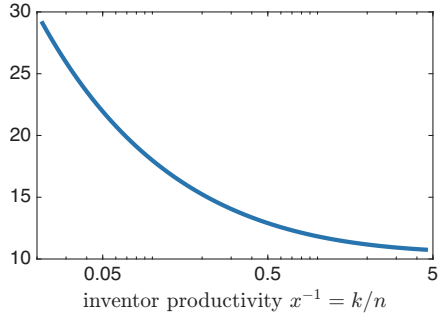
- $S_x(x) = \Omega_n$ and $S(x) - xS_x(x) = \Omega_k$
- If $\alpha = 0$, inventors move only to firms with higher inventor value S_x, as in a job ladder
- With $\alpha > 0$, a move to a firm with lower $x^{-1} = k/n$ occurs when the value of the knowledge brought exceeds the loss in inventor value

Less productive firms value inventors less but knowledge more

Inventor value $S_x(x) = \Omega_n$



Knowledge value $S(x) - xS_x(x) = \Omega_k$



Marginal values by β

Mobility shapes growth through reallocation and diffusion

- A Cobb–Douglas aggregator and a quality ladder with step λ give growth

$$g = \underbrace{\log(\lambda)}_{\text{innovation step}} \times \underbrace{(\delta_{RD} + \delta_{KD})}_{\text{creative destruction rate}}$$

$$\underbrace{\delta_{RD}}_{\text{internal R\&D}} = \int \mu(x)\, dF(x), \qquad \underbrace{\delta_{KD}}_{\text{knowledge diffusion}} = N\theta q(\theta) \iint \mathbb{1}_P(x, x')\, \alpha(x')\, dF_x(x')\, dF_v(x)$$

- $\mathbb{1}_P(x, x')$: equals 1 when the match surplus $M(x, x')$ is positive
- Since μ is concave, δ_{RD} is highest when inventor productivity is equal across product lines

Sources of Private-Social Wedges

Private-social wedge	Source	Effect on private incentives
Innovation value	Appropriability	Too little private innovation value
Innovation value	Business stealing	Too much private innovation value
Vacancy posting	Hold-up from surplus sharing	Too few vacancies
Vacancy posting	Search congestion	Too many vacancies

Estimation

What INV-BIO Observes

- **Inventors.** Establishment and dates of each job, daily wage, occupation, education, and age
- **Patents.** Filing date, forward citation counts, co-inventors, and technology classes
- **Establishments.** Employment, mean wage, industry, and region

Functional Forms

Function	Description
$c(v) = \frac{\bar{c}}{\phi+1} v^{\phi+1}$	Vacancy cost function
$q(\theta) = \bar{q}\theta^{\eta-1}$	Matching function
$\mu(x) = \bar{\mu} x^{1/2}$	Internal R&D function
$\alpha(x') = \bar{\alpha} x'^{-1}$	Knowledge diffusion function

From the Law of Motion to a Regression

- e: establishment; t: year
- Inventor productivity $z \equiv x^{-1} = k/n$ is measured by $z_{e,t}$, three-year citations per inventor at e to their patents filed in t
- $\alpha(x') = \overline{\alpha}\, z'$ and $\mu(x) = \overline{\mu}\, z^{-1/2}$
- $z^{\text{inc}}_{e,t}$ is the productivity of inventors who stay at e from $t-1$ to t
- $m_{e' \to e,t}$ inventors move from e' to e between $t-1$ and t

$$\Delta \log z^{\text{inc}}_{e,t} = \underbrace{\overline{\alpha} \sum_{e'} \frac{m_{e' \to e,t}}{n_{e,t-1}} \frac{z_{e',t-1}}{z_{e,t-1}}}_{\text{diffusion exposure}} + 12\,\overline{\mu} \underbrace{z_{e,t-1}^{-1/2}}_{\substack{\text{R\&D} \\ \text{exposure}}} + \underbrace{\text{net inflow rate} + \text{FE}}_{\text{controls}} + \varepsilon_{e,t}$$

Estimates of Knowledge Diffusion and Internal R&D

	Incumbent productivity growth		
	(1)	(2)	(3)
Diffusion exposure	0.526***	0.306***	0.250***
	(0.071)	(0.073)	(0.089)
R&D exposure $z_{e,t-1}^{-1/2}$	0.255***	0.347***	0.440***
	(0.023)	(0.024)	(0.029)
Establishment FE	Yes	Yes	Yes
Year FE	Yes	No	No
Industry × region × year FE	No	Yes	Yes
Establishment linear trend	No	No	Yes
Root MSE	0.702	0.641	0.653
Establishment-years	12,690	10,812	10,812

- Column (1), the model's moment net of establishment and year effects, sets

$$\overline{\alpha} = 0.526$$
$$\overline{\mu} = 0.2546/12 = 0.0212$$
$$\sigma = 0.702/\sqrt{12} = 0.203$$

INV-BIO establishment-years with positive exposure, weighted by incumbent inventors; net inflow rate in all columns; standard errors clustered by establishment. Annual data, monthly model.

Robustness table Source vs. mover

Incumbent productivity rises with predicted source-contraction inflows

- Predicted moves allocate inventors at severely contracting sources in proportion to each source's moves two to six years earlier

$$\hat{m}_{e' \to e,t} = \underbrace{s_{e' \to e,t}}_{\substack{\text{share of } e'\text{'s} \\ \text{past moves to } e}} \times \underbrace{n_{e',t-1} \, \mathbb{1}\{\text{employment at } e' \text{ falls by at least 80\%}\}}_{\text{displaced inventors}}$$

- Predicted exposure is the diffusion exposure with $\hat{m}_{e' \to e,t}$ in place of $m_{e' \to e,t}$

Coefficient on predicted exposure	(1)	(2)	(3)
First stage			
Realized diffusion exposure	0.0236**	0.0245*	0.0363
	(0.0117)	(0.0125)	(0.0231)
Reduced form			
Incumbent productivity growth	0.0515**	0.0482**	0.0449
	(0.0200)	(0.0187)	(0.0295)

INV-BIO establishment-years; fixed effects as in baseline columns (1)–(3); standard errors clustered by establishment.

Calibration

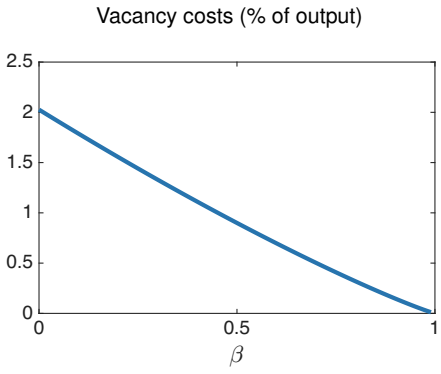
Set externally or normalized		
ρ	0.004	Monthly discount rate (5% a year)
$\overline{c}$	1	Vacancy-cost scale, normalized
ϕ	3.45	Vacancy-cost curvature, Bilal et al. (2022)
η	0.50	Matching elasticity, Petrongolo and Pissarides (2001)
ε	0.50	R&D elasticity, Acemoglu et al. (2018)
π	0.23	Profit per product line (markup 1.3)
N	1.95	Inventors ($1/N = 0.512$ citations)
β	0.50	Baseline surplus share

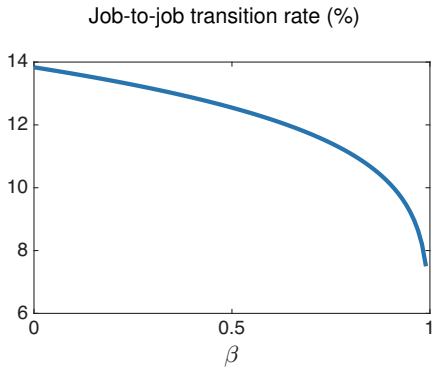
Regression, column (1)		
$\overline{\alpha}$	0.526	Knowledge diffusion
$\overline{\mu}$	0.0212	Internal R&D
σ	0.203	R&D shocks
Balanced-growth targets		
$\overline{q}$	0.0198	Job-to-job rate 12.5% (2001)
λ	1.053	Growth 2% a year

- With $\beta = 0.5$, a ban raises the job-to-job rate by 10.2%, close to Hawaii's 11% rise (Balasubramanian et al., 2022)

Quantitative Exercises

Stronger non-compete enforcement lowers recruiting and mobility

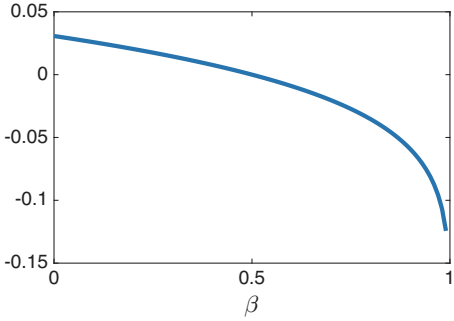




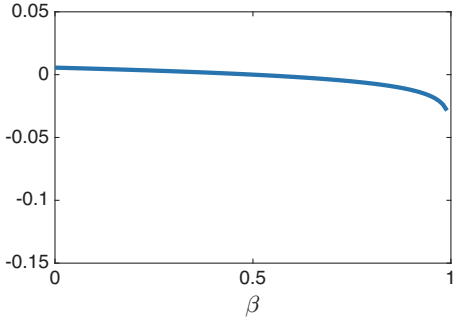
- Steady states by β

Stronger enforcement lowers growth, mostly through diffusion

Knowledge-diffusion growth, change from $\beta = 0.5$ (pp)



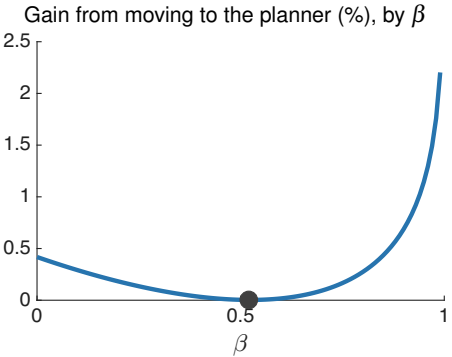
Internal-R&D growth, change from $\beta = 0.5$ (pp)



Steady states, annual rates on a common vertical scale.

- At $\beta = 0.5$, knowledge diffusion accounts for 16% of growth
- However, most of the change in growth comes from knowledge diffusion
- Internal-R&D growth responds only through productivity dispersion, which moves little

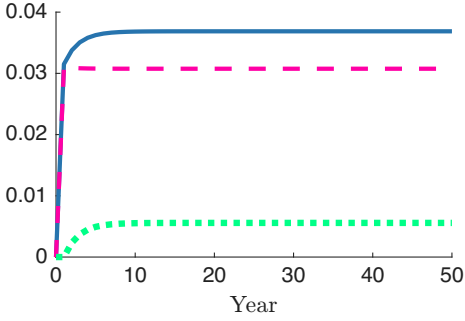
A fixed split nearly implements the planner



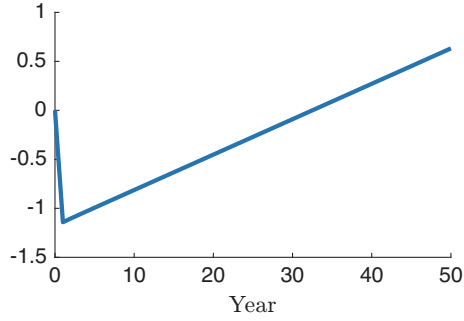
- For each β, the figure shows the CE gain from moving from its steady state to the planner
- The gain is smallest at $\beta^* = 0.52$, at 0.0029% of consumption
- A ban raises welfare only if β exceeds 0.85 before the ban

A ban raises growth and lowers welfare

Growth (pp): total, diffusion (dashed), R&D (dotted)



Log consumption, change from initial BGP ($\times 100$)

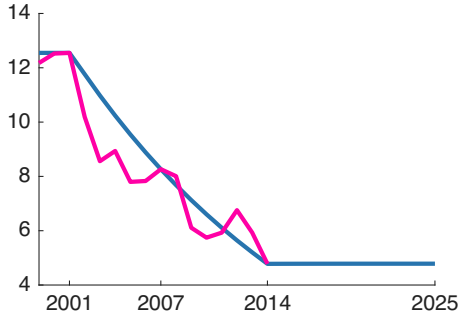


- β falls permanently from 0.5 to 0, and the job-to-job rate rises by 10.2%
- Long-run growth rises by 0.037 pp, 0.031 pp of it through knowledge diffusion
- Firms spend more on recruiting, so CE welfare falls by 0.41%

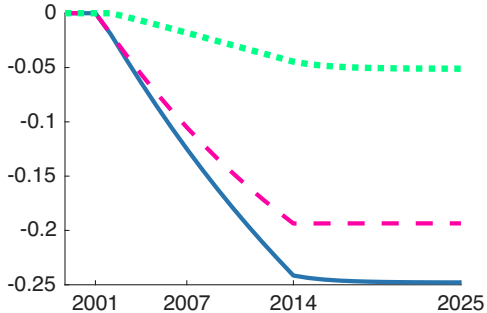
All transition panels Welfare measure

Declining inventor mobility lowers growth and welfare

Job-to-job rate (%): model (blue), data (magenta)



Growth (pp): total, diffusion (dashed), R&D (dotted)



- A path of matching efficiency matches the rates in 2001, 2007, and 2014 (12.5%, 8.3%, 4.8%)
- Long-run growth falls by 0.25 pp, 0.19 pp of it through knowledge diffusion
- CE welfare falls by 3.46%

Matching-efficiency path Annual mobility rates

Conclusion

- Inventors often move to lower-wage establishments yet gain more in wages than other movers
- A fixed split nearly implements the planner ($\beta^* = 0.52$)
- From $\beta = 0.5$, a non-compete ban raises growth and lowers welfare by 0.41%
- The 2001–2014 mobility decline lowers annual growth by 0.25 pp and welfare by 3.46%
- Knowledge diffusion, not reallocation, drives most of both growth effects

Appendix

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- SIAB and inventor proxies
- Fact 2 specification

Theoretical Appendix

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- Aggregation setting
- BGP equilibrium
- Social optimum and planner's problem
- Decomposition of firm growth

Quantitative Appendix

- Source vs. mover
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- Model fit
- Steady states
- Ban welfare by β
- Matching elasticity
- Transitions and welfare
- Mobility decline

Summary Statistics: INV-BIO

Establishment-level variables	Mean	S.D.	N, thousand
N of inventors (n_{et})	5.18	29.53	678.9
N of employees	300.76	1088.36	678.9
Mean daily wage, Euro	100.83	54.63	670.3
Three-year backward average citation productivity ($\bar{z}_{et}^{(3)}$)	0.13	0.66	465.6
Total inventor inflows	0.40	2.99	678.9
Inventor productivity growth	-0.02	1.09	13.9
Knowledge-diffusion inflow term	0.16	0.34	13.9
Net inventor inflow rate	0.06	0.14	13.9

$$z_{et} = \frac{1}{|J_{et}|} \sum_{j \in J_{et}} \sum_{i \in I_{jt}} \frac{\text{three-year citations}_i}{\text{inventors}_i}$$

J_{et}: inventors at e in year t; I_{jt}: patents filed by inventor j in year t.

Back to data Back to regression

Fact 1: Moves with Own-Wage Gains

Ranked by citation productivity (row %)

	Destination rank bin									
Origin	1	2	3	4	5	6	7	8	9	10
1	12	11	10	10	8	9	9	10	10	10
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4	11	10	11	11	10	10	11	9	9	8
5	8	8	9	8	9	11	16	10	9	10
6	7	8	10	7	11	10	10	12	11	14
7	5	7	6	14	12	9	12	11	11	12
8	5	6	7	6	8	13	12	15	15	14
9	4	5	11	6	10	8	9	17	12	18
10	4	7	5	6	5	10	10	11	16	26

Ranked by mean wage (row %)

	Destination rank bin									
Origin	1	2	3	4	5	6	7	8	9	10
1	17	16	13	12	10	9	7	7	6	4
2	10	16	13	12	10	9	8	8	7	6
3	7	12	13	12	10	9	9	9	9	9
4	5	9	11	11	11	11	11	11	11	9
5	5	8	9	11	10	11	10	12	12	13
6	3	7	8	9	10	13	13	12	13	12
7	3	6	7	8	9	11	13	13	15	16
8	2	5	5	7	10	11	14	14	15	18
9	2	4	5	5	9	9	14	14	16	22
10	2	3	4	4	5	7	11	12	18	34

INV-BIO, 1980–2014, inventor job-to-job moves in which the inventor's own wage rises.

Back to Fact 1

SIAB Data: Summary Statistics

Worker-level variables	Mean	S.D.	N, thousand
D_{it}^{wage}	0.45	–	652
$\Delta \log w_{it}$	0.173	0.657	652
Wage increase indicator	0.72	–	652
$I_{i,t-1}$	0.10	–	652
Age	33.3	11.3	652
Female, %	42.7	–	652
University-entrance qualification, %	27.0	–	652
Log wage	3.877	0.871	652

Back to data

Definition of Inventor-Proxy Workers

- Inventor-proxy workers: SIAB workers with a university-entrance qualification in the five largest KldB 2010 parent occupation groups among INV-BIO inventors
  - Machine and vehicle technology (25)
  - Mechatronics, energy and electrical engineering (26)
  - Technical R&D, design and production planning (27)
  - Mathematics, biology, chemistry and physics (41)
  - Business management and organisation (71)
- These groups account for 80.5% of inventor-years with observed occupations

Inventor-Proxy Workers in SIAB and Inventors in INV-BIO

Summary statistics (1980–2014)		SIAB		INV-BIO
		Workers	Inventor-proxy workers	Inventors
Wage, Euro	Mean	63.8	99.5	123.8
	S.D.	45.5	55.6	52.1
Age	Mean	38.5	39.0	38.9
	S.D.	12.6	11.2	10.6
Females, %		44.2	33.7	6.4
University-entrance qualification, %		15.3	100.0	70.3
N of observations, thousand		17,474	1,030	3,517

Inventor-proxy workers are workers whose observed KldB 2010 parent occupation group is 25, 26, 27, 41, or 71 and who have a university-entrance qualification.

Fact 2: Regression Specification

- Compare inventor-proxy workers with other job-to-job movers

$$y_{it} = b\, l_{i,t-1} + X'_{i,t-1}\gamma + \mathrm{FE}_{o(i)} + \mathrm{FE}_{d(i)} + \mathrm{FE}_t + \varepsilon_{it}$$

- y_{it}: lower-mean-wage destination or wage growth
- Sample: BeH-to-BeH establishment changes without unemployment spell
- $X_{i,t-1}$: age, age squared, gender, and log wage

Back to Fact 2

Inventor-Proxy Worker Flows: Wage-Increase Moves

	Wage-increase job-to-job sample	
	(1)	(2)
Dependent variable	D_{it}^{wage}	$\Delta \log w_{it}$
$I_{i,t-1}$	0.0094***	0.0535***
	(0.0025)	(0.0027)
Controls	Yes	Yes
Origin estab. FE	Yes	Yes
Dest. estab. FE	Yes	Yes
Year FE	Yes	Yes
Observations	404,722	404,722
Adjusted R^2	0.5678	0.7002

Sample is restricted to establishment changes with an increase in the worker's own daily wage. Other definitions match the full job-to-job sample.

Back to Fact 2

Inventor Mass per Product Line x as State Variable

- $x \equiv n/k$: inventor mass per product line
- $x^{-1} = k/n$: inventor productivity
- $f(x)$: distribution of product lines over x
- $S(x)$: joint value of x mass of inventors and unit mass of product line

$$\underbrace{kg(k,n)}_{\text{\# of product line} \times \text{Firm distribution}} = \underbrace{f(x)}_{\text{Product distribution}}$$

$$\underbrace{\Omega(k,n)}_{\text{Firm-level joint value}} = \underbrace{kS(x)}_{\text{\# of product line} \times \text{Product-level joint value}}$$

$$\underbrace{\Omega_n(k,n)}_{\text{Marginal value of inventor}} = \underbrace{S_x(x)}_{\text{Inventor per product line} \uparrow}$$

$$\underbrace{\Omega_k(k,n)}_{\text{Marginal value of knowledge capital}} = \underbrace{S(x)}_{\text{Product line} \uparrow} \quad \underbrace{-xS_x(x)}_{\text{Inventor per product line} \downarrow}$$

Back to simplified HJB

Aggregation Setting

- Representative household utility

$$\int_0^\infty e^{-\rho t} \log C(t) dt$$

- Inelastically supplies L units of production labor and N inventors
- Aggregate output

$$\log Y(t) = \int_0^1 \log \left(a(\omega, t) l(\omega, t) \right) d\omega$$

  - $a(\omega, t)$: productivity of product ω
  - $l(\omega, t)$: production labor used to produce product ω (no friction)

- Arrival of creative destruction is random across product lines
- Innovation step size is $\lambda > 0$

BGP Equilibrium 1

A stationary competitive equilibrium consists of sequence of joint value, vacancy policy, poaching indicator function, inventor productivity distribution, creative destruction rate, and labor market tightness $\left[\left\{S(x), v(x), \{\mathbb{1}_P(x,x')\}_{x'\in(0,\infty)}, f(x)\right\}_{x\in(0,\infty)}, \delta, \theta\right]$ such that

- The joint value $S(x)$ satisfies the HJB equation

$$\begin{aligned}\rho S(x) =& \pi - c(v(x)) + \{\mu(x) - \delta\}\{S(x) - xS_x(x)\} + \frac{\sigma^2}{2}x^2 S_{xx}(x) \\ &+ (1-\beta)q(\theta)v(x)\int M(x,x')dF_x(x') + \beta\theta q(\theta)x\int M(x',x)dF_v(x')\end{aligned}$$

where the match surplus $M(x,x')$ is given by

$$M(x,x') = \left[S_x(x) + \alpha(x')\{S(x) - xS_x(x)\} - S_x(x')\right]^+$$

BGP Equilibrium 2

- The vacancy posting is optimal, so that $v(x,t)$ satisfies

$$c'(v(x)) = (1-\beta)\, q(\theta) \int M(x,x')dF_x(x')$$

- Given a match, the inventor moves if the match surplus is positive:

$$\mathbb{1}_P(x,x') = \begin{cases} 1 & \text{if} \quad M(x,x') > 0 \\ 0 & \text{otherwise} \end{cases}$$

BGP Equilibrium 3

- The inventor productivity distribution $f(x,t)$ follows the KF equation

$$
\begin{aligned}
0 = & \left\{ \begin{array}{c} \mu(x) - \delta \\ +q(\theta) v(x) \int \mathbb{1}_P(x,x') \alpha(x') dF_x(x') \end{array} \right\} f(x) \\
& - \frac{\partial}{\partial x} \left[\left\{ \begin{array}{c} -\{\mu(x) - \delta\} x \\ +q(\theta) v(x) \int \mathbb{1}_P(x,x') \{1 - \alpha(x') x\} dF_x(x') \\ -\theta q(\theta) x \int \mathbb{1}_P(x',x) dF_v(x') \end{array} \right\} f(x) \right] \\
& + \frac{\partial^2}{\partial x^2} \left[\frac{\sigma^2}{2} x^2 f(x) \right],
\end{aligned}
$$

- The labor market tightness θ and the creative destruction rate δ satisfy

$$
\begin{aligned}
N\theta &= \int v(x) f(x) dx \\
\delta &= \int \mu(x) dF(x) + N\theta q(\theta) \int \int \mathbb{1}_P(x,x') \alpha(x') dF_x(x') dF_v(x)
\end{aligned}
$$

Social Optimum

$$\rho S^*(x) = \max_{v \geq 0} \underbrace{\pi^*}_{\text{Growth externality}} - c(v(x)) + \{\mu(x) - \delta\}\{S^*(x) - xS_x^*(x)\} + \frac{\sigma^2}{2}x^2 S_{xx}^*(x)$$

$$+ \underbrace{\theta q'(\theta) v(x) \int\int M^*(x, x')\ dF_x(x') dF_v(x)}_{\text{Congestion externality}}$$

$$+ \underbrace{q(\theta) v(x) \int M^*(x, x')\, dF_x(x') + \theta q(\theta) x \int M^*(x', x)\, dF_v(x')}_{\text{Hold-up problem}}$$

- Externality from Schumpeterian growth: appropriability effect + business stealing effect
- Externality from random search: congestion externality + hold-up problem

Social Planner's Problem

$$\mathbf{x}(t) = \left[\log Y(t), \{f(x,t)\}_{x\in(0,\infty)}\right]$$

$$\mathbf{y}(t) = \left[\delta(t), \theta(t), \{v(x,t)\}_{x\in(0,\infty)}, \left\{\mathbb{1}_P(x,x',t)\right\}_{x,x'\in(0,\infty)}\right]$$

$$\boldsymbol{\lambda}_{\mathbf{x}}(t) = \left[\lambda_Y(t), \{\lambda_f(x,t)\}_{x\in(0,\infty)}\right]$$

$$\boldsymbol{\lambda}_{\mathbf{y}}(t) = [\lambda_\delta(t), \lambda_\theta(t)]$$

Back to social optimum

Social Planner's Problem

$$
\begin{aligned}
&\mathscr{H}^{P}\left(\mathbf{x}(t),\mathbf{y}(t),\boldsymbol{\lambda}_{\mathbf{x}}(t),\boldsymbol{\lambda}_{\mathbf{y}}(t)\right)\\
&=\ln Y(t)+\ln\left\{1-\int c\left(v(x,t)\right)dF(x,t)\right\}\\
&\quad+\lambda_{Y}(t)\,\delta(t)\ln\lambda\\
&\quad+\lambda_{\delta}(t)\left[-\delta(t)+\int\left\{\mu(x)+q(\theta(t))v(x,t)\int\mathbb{1}_{P}(x,x',t)\alpha\left(x'\right)dF_{x}(x',t)\right\}dF(x,t)\right]\\
&\quad+\lambda_{\theta}(t)\left\{N\theta(t)-\int_{0}^{\infty}v(x,t)\,f(x,t)dx\right\}\\
&\quad+\int_{0}^{\infty}\lambda_{f}(x,t)\left[\begin{array}{c}\left\{\begin{array}{c}q(\theta(t))v(x,t)\int\mathbb{1}_{P}(x,x',t)\alpha\left(x'\right)dF_{x}(x',t)\\+\mu(x)-\delta(t)\end{array}\right\}f(x,t)\\-\frac{\partial}{\partial x}\left[\left\{\begin{array}{c}q(\theta(t))v(x,t)\int\mathbb{1}_{P}(x,x',t)\left\{1-\alpha\left(x'\right)x\right\}dF_{x}(x',t)\\-\theta(t)\,q(\theta(t))x\int\mathbb{1}_{P}(x',x,t)dF_{v}(x',t)\\-\left\{\mu(x)-\delta(t)\right\}x\end{array}\right\}f(x,t)\right]\\+\frac{\partial^{2}}{\partial x^{2}}\left[\frac{\sigma^{2}}{2}x^{2}f(x,t)\right]\end{array}\right]dx
\end{aligned}
$$

Back to social optimum

Social Planner's Problem

$$
\begin{aligned}
0 &= \frac{\partial}{\partial \mathbf{y}(t)} \mathscr{H}^P\left(\mathbf{x}(t), \mathbf{y}(t), \boldsymbol{\lambda}_{\mathbf{x}}(t), \boldsymbol{\lambda}_{\mathbf{y}}(t)\right) \\
\rho \boldsymbol{\lambda}_{\mathbf{x}}(t) - \dot{\boldsymbol{\lambda}}_{\mathbf{x}}(t) &= \frac{\partial}{\partial \mathbf{x}(t)} \mathscr{H}^P\left(\mathbf{x}(t), \mathbf{y}(t), \boldsymbol{\lambda}_{\mathbf{x}}(t), \boldsymbol{\lambda}_{\mathbf{y}}(t)\right) \\
0 &= \frac{\partial}{\partial \boldsymbol{\lambda}_{\mathbf{y}}(t)} \mathscr{H}^P\left(\mathbf{x}(t), \mathbf{y}(t), \boldsymbol{\lambda}_{\mathbf{x}}(t), \boldsymbol{\lambda}_{\mathbf{y}}(t)\right) \\
\dot{\mathbf{x}}(t) &= \frac{\partial}{\partial \boldsymbol{\lambda}_{\mathbf{x}}(t)} \mathscr{H}^P\left(\mathbf{x}(t), \mathbf{y}(t), \boldsymbol{\lambda}_{\mathbf{x}}(t), \boldsymbol{\lambda}_{\mathbf{y}}(t)\right) \\
0 &= \lim_{t \to \infty} \left[\exp\left(-\rho t\right) \boldsymbol{\lambda}_{\mathbf{x}}(t) \mathbf{x}(t)\right]
\end{aligned}
$$

Back to social optimum

Optimal Instantaneous Profit

$$\underbrace{\frac{\ln \lambda}{\rho}\left[1-\int_0^{\infty} c(v(x))\, f(x) dx\right]}_{\text{marginal gain from innovation}} = \underbrace{\int_0^{\infty}\left[S^*(x)-S_x^*(x)\, x\right] f(x) dx}_{\text{expected marginal value of knowledge}}$$

Back to social optimum

Decomposition of Firm Growth

$$d\log x^{-1}(x,t) = d\log k(x,t) - d\log n(x,t)$$

$$d\log k(x,t) = \underbrace{q(\theta(t))v(x,t)\int \mathbb{1}_P(x,x',t)\alpha(x')\,dF_x(x',t)dt}_{\text{Knowledge Gain from Poaching}} + \underbrace{\left(\mu(x) - \delta(t) - \frac{\sigma^2}{2}\right)dt + \sigma dW(t)}_{\text{R\&D}-\text{Creative Destruction}}$$

$$d\log n(x,t) = \underbrace{\frac{q(\theta(t))v(x,t)}{x}\int \mathbb{1}_P(x,x',t)dF_x(x',t)dt}_{\text{inventor inflow rate}} - \underbrace{\theta(t)\,q(\theta(t))\int \mathbb{1}_P(x',x,t)dF_v(x',t)dt}_{\text{inventor outflow rate}}$$

Back to the regression

The source's productivity, not the mover's, drives the estimate

	Incumbent productivity growth		
	(1)	(2)	(3)
Source productivity, leave-one-mover-out	0.478***	0.273***	0.210**
	(0.070)	(0.080)	(0.099)
Mover's own productivity	0.048	0.040	0.036
	(0.030)	(0.039)	(0.049)
Establishment-years	12,628	10,757	10,757

INV-BIO establishment-years; fixed effects as in baseline columns (1)–(3); standard errors clustered by establishment.

- Without the mover, the source coefficient stays near the baseline estimate, while the mover's own coefficient is small

Back to estimate

Robustness of the Baseline Regression

	Zero-exposure sample			Lagged-outcome control	
	(1)	(2)	(3)	(4)	(5)
Diffusion exposure	0.3151^{***}	0.1803^{***}	0.1213^{**}	0.3127^{***}	0.2341^{**}
	(0.0519)	(0.0514)	(0.0613)	(0.0811)	(0.1006)
Lagged incumbent productivity growth				-0.1982^{***}	-0.1896^{***}
				(0.0180)	(0.0218)
Establishment FE	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	No	No	No	No
Industry $\times$ region $\times$ year FE	No	Yes	Yes	Yes	Yes
Establishment-specific linear trend	No	No	Yes	No	Yes
Establishment-years	32,588	29,907	29,907	8,860	8,860

Columns (1)–(3) add establishment-years with zero exposure. All columns include the R&D exposure and the net inflow rate.

Back to estimate

Untargeted Productivity Dispersion

Moment	Data	Model
SD of log positive inventor productivity	1.08	0.89
P90/P50 of positive inventor productivity	2.94	3.24
P50/P10 of positive inventor productivity	4.81	3.15

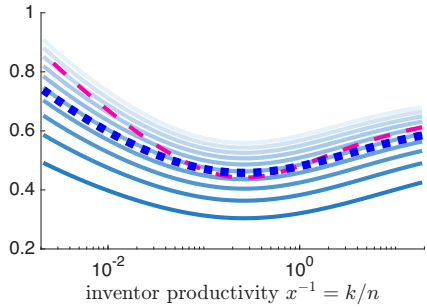
- The model understates overall and lower-tail dispersion and overstates the upper-tail ratio

Data: establishment-years with positive three-year citation productivity, 2001–2011, weighted by inventors. Model: inventor-weighted stationary distribution of k/n.

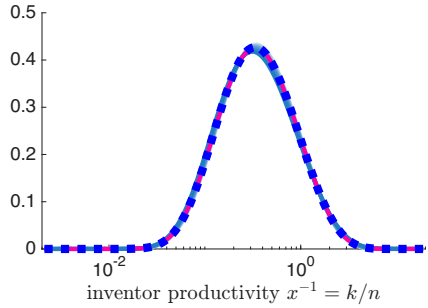
Back to calibration

Vacancy Posting Policy and Distribution

Vacancy posting policy $v(x)$



Inventor-weighted distribution $f_x(x)$

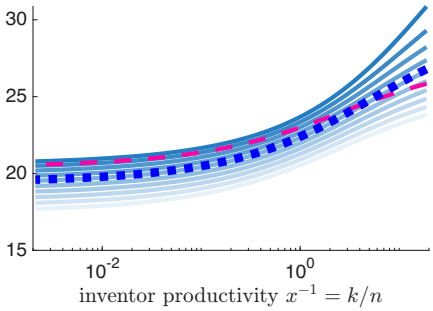


- $\beta \uparrow$ (The share of match surplus of poaching firms $\downarrow$) $\rightarrow$ Vacancy posting $\downarrow$

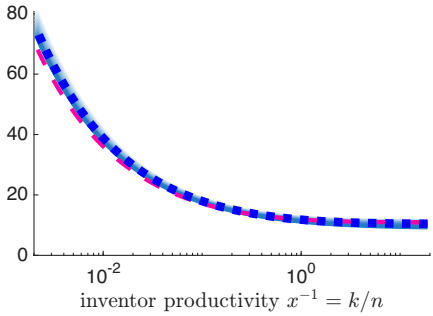
Horizontal axis: inventor productivity $x^{-1} = k/n$. Darker curves: larger $\beta = 0, 0.1, \ldots, 0.9$. Dashed: planner. Dotted: efficient split $\beta^* = 0.52$.

Marginal Value of Inventor & Marginal Value of Knowledge

$\Omega_n(k,n) = S_x(x)$



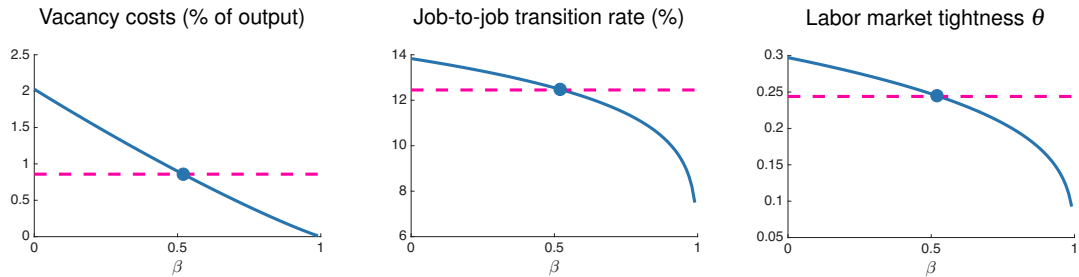
$\Omega_k(k,n) = S(x) - xS_x(x)$



Horizontal axis: inventor productivity $x^{-1} = k/n$. Darker curves: larger $\beta = 0, 0.1, \ldots, 0.9$. Dashed: planner. Dotted: efficient split $\beta^* = 0.52$.

Back to policy functions

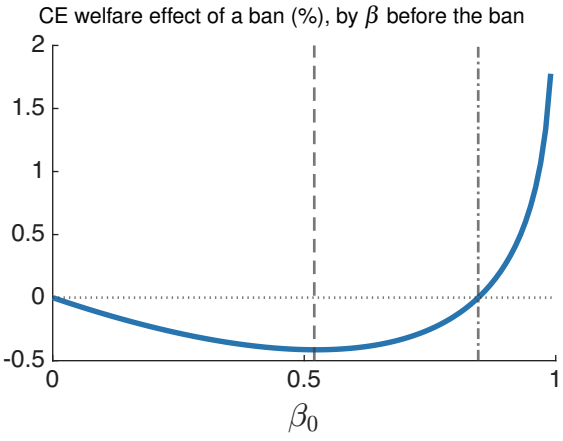
Vacancy Costs, Job-to-Job Rate, and Tightness



Steady states by β. Solid: equilibrium; dashed: planner; dot: efficient split $\beta^* = 0.52$.

Back to steady states

Welfare Effect of a Ban by Pre-Ban Surplus Share



- Every ban leads to the same BGP, so the welfare change tracks pre-ban welfare
- The loss is largest near the efficient split $\beta^* = 0.52$ (dashed line)
- A ban raises welfare only when β exceeds 0.85 before the ban (dash-dotted line)

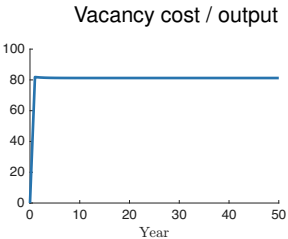
Back to fixed split

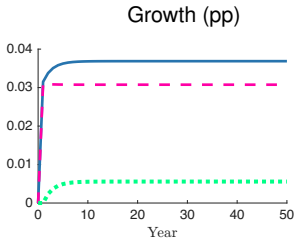
Efficient Split and Ban Welfare by Matching Elasticity

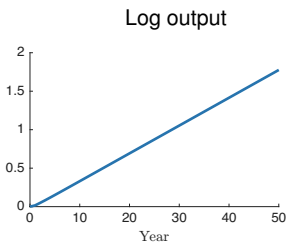
η	$1-\eta$	β^*	Gain at β^* (%)	Welfare change from ban (%)	Threshold β_0
0.3	0.70	0.71	0.0136	-0.66	0.98
0.4	0.60	0.62	0.0065	-0.54	0.93
0.5	0.50	0.52	0.0029	-0.41	0.85
0.6	0.40	0.43	0.0010	-0.28	0.74
0.7	0.30	0.34	0.0002	-0.13	0.62

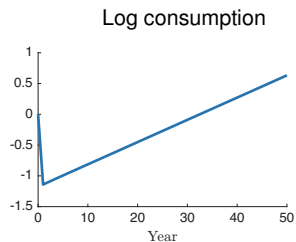
- $\overline{q}$ and λ are recalibrated for each η. β^* stays within 0.04 of $1-\eta$, and the ban lowers welfare at $\beta = 0.5$ for every η

Transition after a Ban









- Vacancy costs jump on impact; growth rises mainly through knowledge diffusion
- Output rises gradually; consumption falls on impact before recovering
- The growth gain from a ban rises from 0.037 pp at $\beta_0 = 0.5$ to 0.136 pp at $\beta_0 = 0.95$

Levels: 100 times the log deviation from the initial BGP. Growth: solid, total; dashed, knowledge diffusion; dotted, internal R&D. Horizontal axis: years.

Back to the ban

Consumption-Equivalent Welfare

$$\log C(t) = \log\lambda \int_0^t \delta(\tau)\, d\tau + \log L + \log\Big\{1 - \int c\big(v(x,t)\big)\, dF(x,t)\Big\}$$

$$V = \int_0^\infty e^{-\rho t} \log C(t)\, dt, \qquad \mathscr{L} = \exp\big[\rho\,\big(V(\widetilde{C}) - V(C)\big)\big]$$

- C: consumption on the initial BGP; $\widetilde{C}$: consumption after the change. Reported welfare changes are $100(\mathscr{L} - 1)$
- After the simulation horizon, consumption grows at the new steady-state rate

Matching-Efficiency Path

- The path's form follows Akcigit and Ates (2023)

$$\overline{q}_t = \overline{q}_0 + \frac{\exp(-\nu t/T) - 1}{\exp(-\nu) - 1}\left(\overline{q}_T - \overline{q}_0\right), \qquad t \in [0, T]$$

- $t = 0$ is 2001 and $t = T$ is 2014
- $\overline{q}_0 = 0.0198$ from the baseline; $\nu = 0.588$ and $\overline{q}_T = 0.00858$ match 8.3% in 2007 and 4.8% in 2014
- Unanticipated before $t = 0$, perfect foresight after. Matching efficiency stays at $\overline{q}_T$ after 2014 while the economy converges

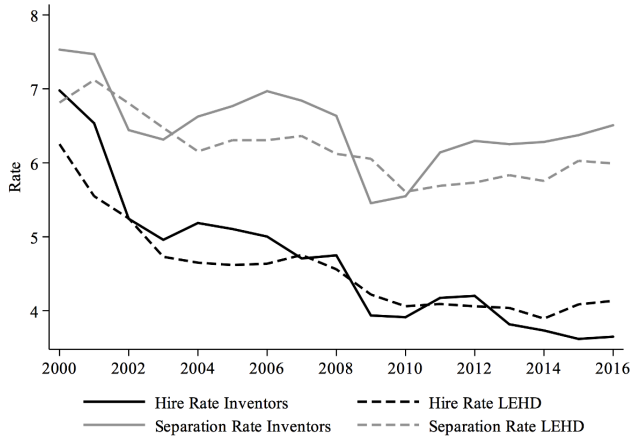
Back to mobility decline

German Inventor Job-to-Job Rates

Year	Inventor person-years	Job-to-job moves	Rate (%)
1999	117,904	14,358	12.2
2000	126,076	15,790	12.5
2001	129,050	16,194	12.5
2002	130,510	13,301	10.2
2003	130,597	11,180	8.6
2004	129,933	11,611	8.9
2005	129,425	10,093	7.8
2006	129,400	10,133	7.8
2007	129,446	10,699	8.3
2008	129,127	10,340	8.0
2009	127,244	7,773	6.1
2010	124,402	7,146	5.7
2011	122,456	7,260	5.9
2012	120,339	8,130	6.8
2013	118,021	6,998	5.9
2014	115,710	5,531	4.8
Total	2,009,640	166,537	8.3

Back to mobility decline

U.S. Inventor Hire and Separation Rates



Akcigit and Goldschlag (2025), based on the Inventor Employment History, with a 1% LEHD sample for comparison.

Back to mobility decline

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